Chapter 8 Answers

Problem 1

Consider some element A. By C_2, there is a mocker M such that $MA = AA$. Thus, by C_1, there is some element C such that $CC = A(MC) = A(CC)$, that is to say, CC is a fixed point of A.

Problem 2

Suppose that C_1 and C_2 hold. Thus, by Problem 1, the mocker M has some fixed point CC such that $M(CC) = CC$. Thus $M(CC) = (CC)(CC) = (M(CC))(M(CC))$, meaning $M(CC)$ is a fixed point of itself.

It can also be seen that any fixed point of the mocker is egocentric. For, given some such fixed point y, $My = y$ and $My = yy$, meaning $yy = y$, and thus y is egocentric.

Problem 3

(a): Suppose that C_1 holds, and that at least one element A is agreeable. Thus for every element B, there is at least one element x on which A and B agree, and for every A and B, there is some C that combines A with B. Now consider any element x. Thus there is some C that combines x with A, meaning, for every element y, $Cy = x(Ay)$. Now consider the y for which A and C agree. Thus $Ay = Cy = x(Ay) = x(Cy)$, meaning Cy is a fixed point of x, as well as Ay.

(b): It is a special case because a mocker is agreeable. For if M is a mocker, then $Mx = xx$ for every element x. Thus for every element x, M and

x agree upon x itself.

Problem 4

Suppose that C_1 holds and C combines A with B, and that C is agreeable. Thus for every element x, $Cx = A(Bx)$ and there is at least one y such that $Cy = xy$. Now consider any element D. We must show that $Ax = Dx$ for at least one x. By the composition condition, there is some C_1 such that, for every element x, $C_1x = D(Bx)$. Thus there is some element y such that $Cy = C_1y$. Hence $A(By) = Cy = C_1y = D(By)$, that is, $A(By) = D(By)$, meaning A and D agree on By.

Problem 5

Assume the composition condition C_1. Consider any elements A, B and C. Thus there is an element C_1 such that $C_1x = B(Cx)$. Hence there is an element C_2 that combines A and C_1, meaning $C_2x = A(C_1x) = A(B(Cx))$. Thus we take E to be C_2.

Problem 6

Suppose conditions C_1 and C_2 hold. Consider any two elements A and B. Thus there is some C such that $Cx = A(Bx)$ for all x, and C has some fixed point y. Thus $A(By) = Cy$ and $B(Cy) = By$, meaning the pair (By, Cy) is a cross point of the pair (A, B).

Problem 7

Suppose an element A has a fixed point x. Thus $Ax = x$, meaning A is compatible with A under the cross point (x, x).

Problem 8

Suppose that C_1 holds and that there is at least one smug element A. Thus there is some pair (x, y) such that $Ax = y$ and $Ay = x$. Thus there is some C such that $Ci = A(Ai)$ for all i. Hence $Cx = A(Ax) = Ay = x$, meaning x is a fixed point for C. It can also be seen that y is a fixed point for C.

Problem 9

Suppose that A was fixated on two distinct elements B and C. Thus $Ax = B = C$ for every element x, that is, $B = C$, contrary to the assumption that B and C are distinct elements. Thus A cannot be fixated on more than one element.

Note: This problem exemplifies the problem that the equal sign is not defined in this chapter. It seems that he is treating x and y being "distinct elements" as the same as it is not the case that $x = y$. He does not mention this, so it is not entirely clear what he means.

Problem 10

(a): Suppose that y is a fixed point of x. This does not necessarily mean $Ax = y$ for every x.

(b): Suppose that x is fixated on y. Then $Ax = y$ for every x, and in particular, $Ay = y$, meaning y is a fixed point of x.

Problem 11

(a): Consider some narcissistic element A, meaning $Ax = x$ for every element x. Thus $AA = A$, meaning it is egocentric (A is it's own fixed point).

(b): Consider some egocentric element A, meaning $AA = A$. This does not necessarily mean $Ax = A$ for every x.

Problem 12

Suppose that A is narcissistic. Then for all elements x and y, $Ax = A = Ay$, meaning the statement is true.

Problem 13

Suppose that A is narcissistic. Thus $Ax = A$ for all x. Now recall that Axy is an abbreviation for $(Ax)y$. Thus $Axy = (Ax)y = Ay = A$ for all x and y.

Problem 14

Suppose that A is narcissistic. By Problem 13, $Axy = A$ for all elements x and y. Thus, because $A = Ax$, Ax is narcissistic for every x.

Problem 15

Suppose that K is a kestrel, and that y is a fixed point of K, meaning $Ky = y$, and $Kxy = x$ for every x and y. Recall that $Ky(Ky) = (Ky)(Ky)$. Thus $yy = (Ky)(Ky) = Ky(Ky) = y$, proving that y is egocentric.

Problem 16

Suppose Kx is egocentric (and K is a kestrel). Thus $x = Kx(Kx) = Kx$, proving that x is a fixed point of K.

Problem 17

Suppose that Kx is a fixed point of K. Thus Kx is egocentric by Problem 15, and x is therefore a fixed point of K by Problem 16.

Problem 18

Suppose that $Kx = Ky$. Thus $x = Kxy = (Kx)y = (Ky)y = Kyy = y$.

Problem 19

Suppose that $Kx = K$ for some x. Now consider any elements y and z. Thus $x = Kxy = (Kx)y = Ky$ and $x = Kxz = (Kx)z = Kz$, meaning $Ky = Kz$. But then, by Problem 18, it follows that $y = z$. Thus it would follow that all elements in $\mathcal{C}$ are the same, which contradicts the fact that $\mathcal{C}$ was assumed to have more than one element.

Note: Note why this problem does not contradict the premise of Problem 17. For if Kx is a fixed point of K, then $K = KKx$, but Kx is not necessarily equal to KKx, as $KKx = (KK)x$, not $K(Kx)$.

Problem 20

(a): This is immediate from Problem 19, taking x to be K itself.

(b): Suppose that y is some fixed point of K. Thus $Ky = y$. Thus, for any element x, $yx = (Ky)x = Kyx = y$, proving that y is narcissistic.

Problem 21

Consider the set $\{K, KK, KKK, ...\}$, and we are considering whether it is infinte. That is, we need to determine how many distinct elements it has. Now, let $K_1 = K$, $K_2 = KK$, and in general, K_n be n K's compounded together. Now, $KKK = K$, $KKKK = ((KK)K)K = (KKK)K = KK$, $KKKKK = (KKKK)K = (KK)K = KKK = K$, and in general, a string of n K's equals K if n is odd, and equals KK if n is even. Here is the proof:

The base cases are given above. Now suppose that the hypothesis holds some n. Thus $K_{n+1} = (K_n)K$. So if $n+1$ is even, n is odd, meaning $(K_n)K = KK$ by the assumption, and if $n+1$ is odd, then n is even, meaning $(K_n)K = (KK)K = K$ by the assumption.

Problem 22

Consider the set $\{K, KK, K(KK), K(K(KK)), ...\}$, that is, the set $\{K_1, K_2, ..., K_n, ...\}$ where $K_1 = K$ and $K_{n+1} = K(K_n)$ for all $n > 1$. Now, by Problem 19, $KK \neq K$. Now we prove by induction that $K_n \neq K_m$ for all $m < n$. We have already proved the base case when $n = 2$, so now suppose that the hypothesis holds for some n. Now, $K_{n+1} = K(K_n)$. By Problem 19, $K_{n+1} \neq K$. Also, if $K(K_n) = K_m$ for some $1 < m < n$, we know that $K(K_n) = K(K_{m-1})$ (as $K_m = K(K_{m-1})$), and thus, by Problem 18, $K_n = K_{m-1}$, a contradiction (as we assumed $K_n \neq K_m$ for all $m < n$). Thus $K_{n+1} \neq K_m$ for all $m < n + 1$, meaning K_{n+1} is distinct from all K_m such that $m < n+1$. Thus, by induction, all elements in the set are distinct, meaning it is an infinite set.

Problem 23

Suppose we have some lark L. Thus $Lxy = x(yy)$. Now consider any element x. Thus $Lx(Lx) = x((Lx)(Lx)) = x(Lx(Lx))$ thus proving that $Lx(Lx)$ is a fixed point of x.

Problem 24

Suppose that some L was both a lark and a kestrel. Thus, for all x and y, $Lxy = x$ and $Lxy = x(yy)$. Hence for all y, $LLy = L$ and $LLy = L(yy)$, meaning $L = L(yy)$, which is not possible by Problem 19.

Problem 25

Suppose that a lark L is narcissistic. Thus, for all elements x and y, $Lxy = x(yy)$ and $Lx = L$. Hence $xL = x(LL) = LxL = (Lx)L = LL = L$ for all x, proving that L is a fixed point of x.

Problem 26

Suppose that some kestrel K was a fixed point of a lark L. Thus $LK = K$, meaning $KK = (LK)K = LKK = K(KK)$, and thus, by Problem

18, $K = KK$, which contradicts Problem 19.

Problem 27

Suppose that some lark L is a fixed point of a kestrel. Thus $KL = L$. Now consider any element x. Hence $L = KLx = Lx$, meaning L is narcissistic, and is therefore a fixed point of x by Problem 25.

Problem 28

Suppose that we are given some lark L. Now, let us show that any fixed point x of LL is such that xx is egocentric. For suppose that x is a fixed point of LL. Thus $L(xx) = LLx = x$, meaning $xx = (L(xx))(L(xx)) = L(xx)(L(xx)) = (xx)(L(xx)L(xx)) = (xx)(xx)$, thus proving that xx is egocentric. Now, by Problem 23, LL does have a fixed point, in particular, the fixed point $L(LL)(L(LL))$. Thus the element $L(LL)(L(LL))(L(LL(L(LL))))$ is egocentric.

Problem 29

(a): Suppose that an identity element I is agreeable. Thus for every x, $Ix = x$ and $Iy = xy$ for some y. But $Iy = y$, meaning $xy = y$, thus proving y is a fixed point of x. On the other hand, suppose that every element has a fixed point. Now consider some elemnt x and its fixed point y. Thus $Iy = y$ and $xy = y$, meaning I agrees with x on y, thus meaning I is agreeable.

(b): Now suppose that every pair of elements is compatible. Now consider any element x. Thus I and x are compatible, meaning for some y and z, $Iy = z$ and $xz = y$. But $Iy = y$, meaning $z = y$. Thus $Iz = z = y$, meaning I and x agree on z.

(c): I is egocentric as $II = I$. But suppose that I was narcissistic. Then $Ix = I$ for every x. But $Ix = x$ for every x. Thus $x = I$ for every x, meaning the only element in $\mathcal{C}$ would be I. But this contradicts the assumption that $\mathcal{C}$ contained more than one element. Hence I cannot be narcissistic.

Problem 30

Suppose that $\mathcal{C}$ contains a lark L and an identity element I. Thus for every y, $LIy = I(yy) = yy$, meaning LI is a mocker.

Problem 31

Suppose that $\mathcal{C}$ contains a lark L and mocker M, and the composition condition holds. By Problem 23, $Lx(Lx)$ is a fixed point of x for every x. Consider the C that combines M with L. Thus $Cx = M(Lx)$. Hence $x(Cx) = x(M(Lx)) = x(Lx(Lx)) = Lx(Lx) = M(Lx) = Cx$, proving that C is a sage.